1. Find out the list of companies which are conducting the placement drive using DSA.

2. Find out the syllabus of GATE for DSA.

③ find out set of inputs and outputs for the following:
(i) Electricity bill
(ii) Bus ticket
(iii) Message to your friend.

④ Today when you started from room to college what are the data structures you identified in your surroundings.

① List of companies

product-Based companies

- Google
- Microsoft
- Amazon
- Adobe
- Apple
- oracle
- salesforce
- Atlassian
- uber
- NVIDIA
- Qualcomm
- paypal
- Intuit
- Flipkart
- Meesho
- phonepay
- Razorpay
- Freshworks

service-Base companies

- Tata consultancy services
- Infosys
- Tech mahindra
- IBM
- LTIMindtree
- Wipro
- cognizant
- Accenture
- Capgemini
- HCL Tech.

Finance & consulting

- Goldman Sachs
- JPMorgan chase
- BlackRock
- Deloitte
- EY
- PwC

② GATE DSA syllabus

① programming in c
- Variables and data types
- operators and expressions
- control statements
- Functions
- Arrays
- pointers
- structures and unions
- Dynamic memory allocation
- File handling
- Recursion.

② Data structures:
- Arrays
- Linked lists (singly, Doubly, circular)
- stacks
- Queues (simple, circular, priority, Deque)
- Trees
  - → Binary Trees
  - → Binary search Trees (BST)
  - → AVL Trees (basic)
  - → Binary Heaps
- Graphs
  - → Representation (Adjacency matrix/List)
  - → BFS
  - → DFS
- Hash Tables.

③ Algorithms
- Asymptotic Analysis
  - → Big-O
  - → Big-Ω
  - → Big-Θ
  - → Time and space complexity
- sorting
  - → Bubble sort
  - → selection sort
  - → Insertion sort
  - → Merge sort
  - → Quick sort
  - → Heap sort

- Searching
  - → Linear search
  - → Binary search
- hashing
- Divide and conquer
- Greedy Algorithms
- Dynamic programming
- Graph Algorithms
  - → Minimum spanning Tree (Prim's, kruskal's)
  - → Shortest path (Dijkstra's, Bellman-Ford basics)
  - → Topological sorting
- Recurrence Relations

③ set of inputs and outputs

① Electricity bill

Inputs
- Customer name
- previous meter reading
- current meter reading
- Rate per unit
- Customer ID

outputs
- Customer ID
- Customer Name
- units consumed
- Total electricity bill.

② Message to your friend

Inputs
- Friend's Name
- Message

outputs
- The message displayed with the friend's name.

③ Bus ticket

Inputs:
- Passenger Name
- source
- Destination
- no. of Tickets
- Fare per ticket.

output:

- passenger Name
- source
- Destination
- no of tickets
- Total fare
- Ticket details

④ data structures in our surroundings

- students waiting for bus → Queue
- Books and bags in hostel → stack
- students walking in a line → linked list
- Vechicles parked in row → Array
- Trees on the campus → Trees
- Roads on the campus → Graph
- college attandence → linked list
- contacts in phone → Array.